Engineering Mathematics (AHT-003) - Unit 5

Unit 5 - Matrices - notes in easy English, matched to the UTU syllabus.

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What this unit covers (syllabus): matrices and their types, rank, consistency of linear systems, eigen values and eigen vectors, Cayley-Hamilton theorem, diagonalisation.

Key ideas: Know the types: symmetric ($A = A^T$), skew-symmetric ($A = -A^T$, diagonal zeros), orthogonal ($AA^T = I$), identity, singular ($|A| = 0$). **Rank** = number of non-zero rows after reducing to echelon form. For a system $Ax = b$: consistent (unique solution) when rank(A) = rank([A|b]) = number of unknowns; infinitely many when both ranks are equal but smaller; **inconsistent** when rank(A) < rank([A|b]). **Eigen values** come from $|A - \lambda I| = 0$ (characteristic equation); for each λ, solve $(A - \lambda I)x = 0$ for the **eigen vector**. **Cayley-Hamilton:** every matrix satisfies its own characteristic equation - use it to find A^{-1} and higher powers. Properties: sum of eigen values = trace, product = determinant.

Exam pointers: the big repeated questions: find eigen values and vectors of a 3x3, verify Cayley-Hamilton and compute A^{-1} from it, and test a system for consistency using ranks. Trace/determinant cross-checks catch arithmetic slips - always do them.